A Priori Error Analysis

Convergence rates, duality, and numerical verification

Lecture outline

1. Identify how polynomial degree and solution regularity limit the predicted convergence rates.
2. Derive the optimal L^2 estimate using a dual problem.
3. Compare predicted rates with errors computed on a mesh sequence.
4. State the corresponding conclusions for linear elasticity.

Convergence rates and solution regularity

Determine how the polynomial degree and the smoothness of the exact solution limit the predicted rate.

Starting point from the previous lecture

For $u \in H^s(\Omega)$ with $2 \leq s \leq p+1$, the previous lecture established

$$\|u - u_h\|_a = O(h^{s-1}), \qquad \|u - u_h\|_{H^1(\Omega)} = O(h^{s-1}).$$

The immediate consequence

$$\|u - u_h\|_{L^2(\Omega)} \leq \|u - u_h\|_{H^1(\Omega)}$$

also gives an L^2 bound of order h^{s-1}.
The interpolation error is one order smaller in L^2:

$$\|u - \mathcal{I}_h u\|_{L^2(\Omega)} = O(h^s).$$

The first goal of this lecture is to determine when the finite element solution has this higher L^2 rate.

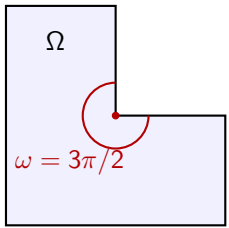
Predicted rates for smooth solutions

Assume $u \in H^{p+1}(\Omega)$. The L^2 rates shown in the last column also require the dual-regularity assumption introduced later.

degree p	energy and H^1 error	L^2 error
1	$O(h)$	$O(h^2)$
2	$O(h^2)$	$O(h^3)$
3	$O(h^3)$	$O(h^4)$

$E(h) = O(h^r)$ means $E(h) \le Ch^r$ with a constant C independent of h. The exponent r is the convergence rate in the selected norm.

Regularity of the exact solution also matters



On a polygon with a reentrant corner, the solution of a smooth-data Laplace problem may fail to belong to $H^2(\Omega)$.

The linear-element estimate requires $u \in H^2(\Omega)$. If this regularity is unavailable, the theorem does not guarantee the usual first-order energy convergence.

Raising the polynomial degree does not repair missing solution regularity. Local refinement near the corner is often more useful; Chapter 8 develops a way to decide where refinement is needed.

In-class calculation: predict the energy rate

Assume the hypotheses of the energy-error theorem. Let s_u be the largest integer for which $u \in H^{s_u}(\Omega)$ is known. Complete the table using

$$r = \min(p, s_u - 1).$$

	$s_u = 2$	$s_u = 3$	$s_u = 4$
$p = 1$			
$p = 2$			
$p = 3$			

Why does changing from $p = 1$ to $p = 3$ not improve the guaranteed rate when $u \in H^2(\Omega)$ but no additional regularity is known?

The optimal L^2 estimate

A dual problem supplies the additional power of h.

Why Céa's lemma does not give the optimal L^2 rate

The H^1 estimate gives the valid bound

$$\|u - u_h\|_{L^2(\Omega)} \leq \|u - u_h\|_{H^1(\Omega)} = O(h^{s-1}).$$

But one-dimensional examples and the interpolation estimate suggest one additional power of h in L^2.

Céa's lemma controls the error in a norm in which the bilinear form is coercive. For diffusion, that is the energy norm, not the L^2 norm.

We therefore introduce an auxiliary problem that converts the squared L^2 error into a bilinear form involving the finite element error.

Step 1: choose the dual problem from the error

Let $e = u - u_h$. Define the dual solution $z \in \mathcal{V}_0$ by

$$\boxed{a(v, z) = (e, v)_{L^2(\Omega)} \qquad \forall v \in \mathcal{V}_0.}$$

For symmetric diffusion, this is the weak form of

$$-\nabla \cdot (k \nabla z) = e \quad \text{in } \Omega,$$

with homogeneous Dirichlet data on Γ_D and homogeneous flux on Γ_N.

Taking $v = e$ gives

$$\boxed{\|e\|^2_{L^2(\Omega)} = a(e, z).}$$

This identity is the reason for introducing the dual problem.

The extra assumption: dual regularity

Lax–Milgram gives $z \in \mathcal{V}_0$, but the interpolation estimate used next requires $z \in H^2(\Omega)$.

We assume

$$\boxed{z \in H^2(\Omega), \qquad |z|_{H^2(\Omega)} \leq C_{\mathrm{reg}} \, \|e\|_{L^2(\Omega)} .}$$

This is a property of the domain, coefficient, and boundary partition, not of the mesh. It holds for many smooth or convex problems, but it can fail at reentrant corners, across coefficient jumps, or where the boundary-condition type changes.

Step 2: use Galerkin orthogonality

Since $z \in H^2(\Omega)$, $\mathcal{I}_h z$ is defined and belongs to $\mathcal{V}_{h,0}$. Galerkin orthogonality gives

$$a(e, \mathcal{I}_h z) = 0.$$

Therefore

$$\|e\|^2_{L^2(\Omega)} = a(e, z) = a(e, z - \mathcal{I}_h z).$$

The dual solution itself is not small, but its interpolation error is:

$$|z - \mathcal{I}_h z|_{H^1(\Omega)} \leq C_I h \, |z|_{H^2(\Omega)} .$$

This factor h produces the additional order in the L^2 error.

Step 3: bound the two factors

$$\begin{aligned}
\|e\|_{L^2(\Omega)}^2 &= a(e, z - \mathcal{I}_h z) \\
&\leq k_1 \, |e|_{H^1(\Omega)} \, |z - \mathcal{I}_h z|_{H^1(\Omega)} \\
&\leq k_1 C_I h \, |e|_{H^1(\Omega)} \, |z|_{H^2(\Omega)} \\
&\leq k_1 C_I C_{\mathrm{reg}} h \, |e|_{H^1(\Omega)} \, \|e\|_{L^2(\Omega)} .
\end{aligned}$$

If $e = 0$, the desired estimate is immediate. Otherwise, divide by $\|e\|_{L^2(\Omega)}$ to obtain

$$\|e\|_{L^2(\Omega)} \leq k_1 C_I C_{\mathrm{reg}} h \, |e|_{H^1(\Omega)} .$$

The dual problem, Galerkin orthogonality, interpolation, and dual regularity have supplied one factor of h.

Step 4: insert the H^1 estimate

The H^1 estimate from the previous lecture implies

$$|e|_{H^1(\Omega)} \leq \|e\|_{H^1(\Omega)} \leq C_E h^{s-1} |u|_{H^s(\Omega)}.$$

Substitute this result into the estimate from Step 3:

$$\begin{aligned} \|e\|_{L^2(\Omega)} &\leq Ch\, |e|_{H^1(\Omega)} \\ &\leq Ch \left(C_E h^{s-1} |u|_{H^s(\Omega)} \right) \\ &= CC_E h^s |u|_{H^s(\Omega)}. \end{aligned}$$

Therefore

$$\boxed{\|u - u_h\|_{L^2(\Omega)} = O(h^s).}$$

For a smooth degree-p approximation, this is $O(h^{p+1})$.

Duality argument: four steps

1. Choose the dual right-hand side to be $e = u - u_h$.
2. Use the dual equation to write $\|e\|_{L^2}^2 = a(e, z)$.
3. Use Galerkin orthogonality to replace z by $z - \mathcal{I}_h z$.
4. Use interpolation and H^2 regularity to obtain $\|e\|_{L^2(\Omega)} \leq Ch\,|e|_{H^1(\Omega)}$, then insert the H^1 finite element estimate to obtain the rate h^s.

Without the additional H^2 regularity of z, Step 3 does not supply a full factor of h. The improved L^2 rate is conditional, not automatic.

Numerical verification

Compare predicted rates with errors computed during mesh refinement.

Observed convergence rate

Suppose $E(h) \approx Ch^r$. For two mesh sizes $h_1 > h_2$,

$$\frac{E(h_1)}{E(h_2)} \approx \left(\frac{h_1}{h_2}\right)^r.$$

Solving for r gives

$$\boxed{r_{\text{obs}} = \frac{\log(E(h_1)/E(h_2))}{\log(h_1/h_2)}.}$$

For uniform refinement with $h_2 = h_1/2$,

$$r_{\text{obs}} = \log_2\left(\frac{E(h)}{E(h/2)}\right).$$

An error reduction by factors 2, 4, and 8 corresponds to observed rates 1, 2, and 3, respectively.

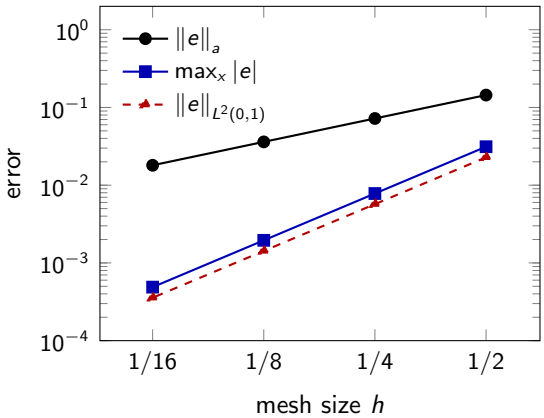
A convergence study tests a specific prediction

A meaningful verification study requires:

1. a known smooth solution, often obtained by the method of manufactured solutions;
2. a mesh sequence satisfying the assumptions of the estimate;
3. the error measured in the norm for which a rate is predicted;
4. sufficiently accurate quadrature for the error integrals; and
5. several refinements in the asymptotic range.

Errors sampled only at the nodes are not sufficient. In some one-dimensional problems, the finite element nodal values are exact even though the error between nodes is nonzero.

Uniformly loaded bar: exact error rates



For $L = AE = f_0 = 1$,

$$u(x) = x - \frac{x^2}{2}, \qquad u_h = \mathcal{I}_h u.$$

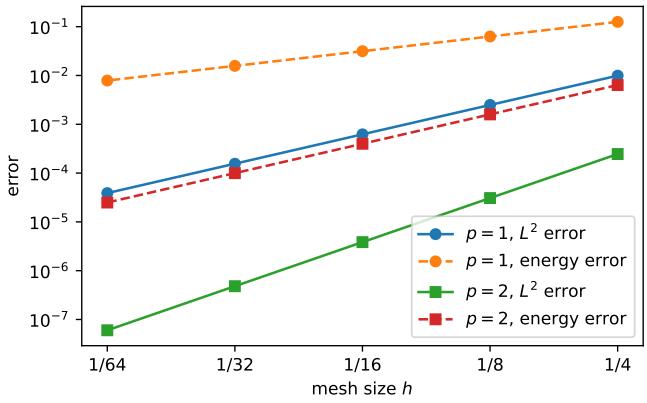
The errors are

$$\max_x |e| = \frac{h^2}{8}, \qquad \|e\|_{L^2(0,1)} = \frac{h^2}{\sqrt{120}},$$

$$\|e\|_a = \frac{h}{\sqrt{12}}.$$

Halving h divides the energy error by 2 and the other errors by 4. The observed rates are 1, 2, and 2.

Manufactured solution: linear and quadratic elements



The manufactured solution is

$$u(x) = \sin \frac{\pi x}{2} \quad \text{on } (0, 1).$$

It is smooth and is not reproduced exactly by either approximation.

	L^2	energy
$p = 1$	2.00	1.00
$p = 2$	3.00	2.00

These are the predicted rates $p + 1$ and p.

In-class calculation: read a two-dimensional study

The Chapter 6 diffusion example used P_1 triangles and produced the following integrated L^2 errors:

triangles	h	L^2 error
4	1.118	1.291×10^{-1}
16	0.559	3.870×10^{-2}
64	0.280	1.079×10^{-2}

1. Compute r_{obs} for each refinement.
2. What rate is predicted for a smooth solution with linear elements?
3. Do two observed rates provide convincing verification? What additional information or computations would you request?

Extension to linear elasticity

Apply the same approximation argument to a vector-valued problem.

The same argument applies to linear elasticity

For homogeneous prescribed displacement data,

$$a_{\mathrm{el}}(\boldsymbol{u} - \boldsymbol{u}_h, \boldsymbol{v}_h) = 0 \qquad \forall \boldsymbol{v}_h \in \mathcal{V}_{h,0}.$$

Continuity, Korn's inequality, and Céa's lemma give

$$\|\boldsymbol{u} - \boldsymbol{u}_h\|_{H^1(\Omega;\mathbb{R}^d)} \leq \frac{C_{\mathbb{C}} C_K^2}{c_{\mathbb{C}}} \inf_{\boldsymbol{v}_h \in \mathcal{V}_{h,0}} \|\boldsymbol{u} - \boldsymbol{v}_h\|_{H^1(\Omega;\mathbb{R}^d)}.$$

Applying the interpolation estimate componentwise gives

$$\|\boldsymbol{u} - \boldsymbol{u}_h\|_{H^1(\Omega;\mathbb{R}^d)} = O(h^{s-1}), \qquad 2 \leq s \leq p+1.$$

Displacement, strain, and stress convergence

If $\boldsymbol{u} \in H^{p+1}(\Omega;\mathbb{R}^d)$, the componentwise interpolation estimate gives

$$\|\boldsymbol{u} - \boldsymbol{u}_h\|_{H^1(\Omega;\mathbb{R}^d)} = O(h^p).$$

The strain and stress errors have the same order because they depend on the first derivatives of displacement.

If the elasticity dual problem satisfies the corresponding H^2 regularity estimate, the duality argument also gives

$$\boxed{\|\boldsymbol{u} - \boldsymbol{u}_h\|_{L^2(\Omega;\mathbb{R}^d)} = O(h^{p+1}).}$$

This improved displacement estimate requires the additional dual-regularity assumption, just as it does for the diffusion problem.

Energy best approximation and compliance

When the elasticity tensor has the major symmetry,

$$\|\boldsymbol{v}\|_{a_{\mathrm{el}}}^2 = \int_\Omega \boldsymbol{\varepsilon}(\boldsymbol{v}) : \mathbb{C} : \boldsymbol{\varepsilon}(\boldsymbol{v}) \, \mathrm{d}\boldsymbol{x}$$

defines the elastic energy norm. The finite element solution is the best approximation in this norm.
For the continuous and discrete solutions,

$$\ell_{\mathrm{el}}(\boldsymbol{u}) = \|\boldsymbol{u}\|_{a_{\mathrm{el}}}^2 , \qquad \ell_{\mathrm{el}}(\boldsymbol{u}_h) = \|\boldsymbol{u}_h\|_{a_{\mathrm{el}}}^2 .$$

The Pythagorean identity gives

$$\boxed{\ell_{\mathrm{el}}(\boldsymbol{u}_h) \leq \ell_{\mathrm{el}}(\boldsymbol{u}).}$$

Thus a conforming displacement approximation underestimates the compliance. This does not order every displacement component at every point.

Material dependence of the elasticity estimate

For a homogeneous isotropic material with $\lambda \geq 0$ and $\mu > 0$, the continuity and coercivity estimates allow us to take

$$C_{\mathbb{C}} = d\lambda + 2\mu, \qquad c_{\mathbb{C}} = 2\mu,$$

so

$$\frac{C_{\mathbb{C}}}{c_{\mathbb{C}}} = 1 + \frac{d\lambda}{2\mu}.$$

For $0 \leq \nu < 1/2$, λ/μ grows without bound as $\nu \to 1/2$. The estimate remains valid, but its constant is not uniform in the material parameters.

Low-order displacement elements may then become excessively stiff, a behavior called **volumetric locking**. Chapter 10 develops mixed formulations for nearly incompressible problems.

Summary

- Shape regularity prevents element geometry from degrading the interpolation constant during refinement.
- Best approximation plus interpolation gives energy and H^1 errors of order h^{s-1}.
- The rate is limited by both polynomial degree and solution regularity.
- Duality gives one additional power of h in L^2 when the dual solution satisfies the required H^2 estimate.
- A convergence study compares predicted and observed rates using integrated error norms over several refinements.
- The same approximation argument applies to conforming linear elasticity, with constants set by Korn's inequality and the material.